

Week 7

1. Focused on improving the reliability and correctness of the queue-based image generation system built in the previous week.
2. Implemented fixes for duplicate job processing issues by analysing worker behaviour and adjusting configurations such as lock duration and concurrency settings in BullMQ.
3. Added explicit job acknowledgement mechanisms to ensure each job is processed exactly once.
4. Developed utility scripts to detect multiple running worker instances and prevent duplicate execution.
5. Improved Redis connection handling and fixed authentication-related issues.
6. Enhanced logging using structured and colored logs for better debugging and traceability.
7. Conducted multiple test runs using producer and consumer scripts to validate that jobs are processed once and status updates are consistent.
8. Overall, stabilised the backend processing pipeline and ensured robustness of the asynchronous system.